

AR MOBILE APP OVERLAYS VIRTUAL OBJECTS ON REAL SCENES

COURSE CODE: DSA0202

COURSE NAME : Computer vision with OPENCV

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INTRODUCTION

What is Augmented Reality (AR)?

AR combines the real world with virtual objects.

- It uses a mobile camera to capture the real scene.
- Virtual objects are placed on the real-world view in real time.

AR is used in games, shopping, education, and entertainment.

Example: Pokémon GO, Snapchat Filters, IKEA Place.

DIFFERENCE BETWEEN COMPUTER GRAPHICS AND COMPUTER VISION

Computer Graphics

Creates virtual objects.

Generates images using a computer.

Output is a virtual scene.

Example: 3D Glasses.

Computer Vision

Understands real-world images.

Analyzes images captured by a camera.

Output is object recognition.

Example: Face Detection.

CAMERA MODEL USAGE

Camera Model

The camera model converts a **3D real-world scene into a 2D image.**

Uses

- Detects camera position.
- Measures object distance.
- Places virtual objects correctly.
- Maintains object alignment while moving.

Example:

A virtual chair stays fixed on the floor when the phone moves.

REAL-TIME CHALLENGES

Challenges in AR

- Fast image processing.
- Accurate object tracking.
- Lighting changes.
- Object occlusion.
- High battery consumption.
- These challenges affect AR performance and accuracy.

AR SYSTEM WORKFLOW

Camera Captures Image



Image Processing



Object Detection
(Computer Vision)



Camera Position Estimation



3D Object Rendering
(Computer Graphics)



AR Display on Mobile Screen

REAL-WORLD APPLICATIONS

- Instagram AR Effects
- Pokémon GO
- IKEA Place
- Google Live View
- Medical AR
- Virtual Try-On Apps

CONCLUSION

- Computer Vision detects and understands real objects.
- Computer Graphics creates virtual objects.
- Camera models ensure accurate object placement.
- Real-time processing provides a smooth AR experience.

AR combines Graphics and Vision to create interactive real-world applications.

THANK YOU !!!